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DEVELOPMENT OF A MACHINE LEARNING MODEL FOR LIVER DISEASE PREDICTION

ABSTRACT

Liver disease is an important health issue worldwide, it was reported that it causes two million death annually. Machine learning (ML) is a relevant technique that is widely used for improving liver disease detection and prognosis. Numbers of studies had explored the use of ML for predicting liver disease based on clinical data. There is need for proper and interpretable predictive models which can be used by the doctors in early detection as well as treatment of diseases. The focus of this research is to develop an Explainable Artificial Intelligence (XAI)-integrated machine learning model for liver disease prediction, addressing key challenges such as class imbalance, interpretability, and clinical usability. The phases of the model development include exploratory analysis, Data preprocessing, training, testing and validation of trained models. Ten machine learning models were trained, tested and evaluated using accuracy, precision, recall, F1-score, and AUC-ROC. The Explainable Boosting Machine (EBM) emerged as the best-performing model with an accuracy of 92.65%, an F1-score of 93.07%, and an AUC-ROC of 0.96. Cross-validation and learning curve analysis confirmed the model's stability and generalization capability. The Explainable AI techniques- SHAP, LIME, and PDP were integrated to increase interpretability and transparency in decision making. The results showed that Alcohol Consumption, Liver Function Test results, and BMI were the most crucial variables in liver disease prediction. Furthermore, a user-friendly application was created to enhance real-time prediction and variable contribution analysis, improving accessibility for healthcare

professionals. The findings of this study demonstrated that integrating XAI with predictive modeling enhances both model reliability and interpretability, making it a valuable tool for clinical decision-making in liver disease diagnosis.

Keywords: Liver Disease; Machine Learning; Explainable AI; Prediction; Model

1.0 Introduction

Liver disease is a major global health concern that causes about two million deaths annually, with males accounting for about two-thirds of these deaths (Devarbhavi et al.,2023). It was reported that in the current century, 4% are due to the liver (Shaheamlun & Kaur, 2021). The first stage of liver illness is hepatitis, it is marked by inflammation and can be caused by viral infections including hepatitis A, B, and C, autoimmune reactions, or alcohol usage. Persistent inflammation may lead to fibrosis, a condition in which scar tissue replaces healthy liver tissue. This progression may culminate in cirrhosis, a severe condition characterized by significant scarring that impairs liver function and can ultimately result in liver failure (Healthline, 2023; Cleveland Clinic, 2024). The causes of liver disease are varied and include chronic alcohol abuse, non-alcoholic fatty liver disease (NAFLD), chronic viral hepatitis, autoimmune hepatitis, genetic (WebMD, 2023; American Liver Foundation, 2024). Machine learning (ML) is a useful technique for improving liver disease detection and prognosis. Numbers of studies have explored the use of ML for predicting liver disease based on clinical data (Mostafa et al., 2021)(Tanwar et al.,2021). These investigations have examined a variety of ML algorithms, such as support vector machines (SVMs), random forests (RFs), and artificial neural networks (ANNs). However, the effectiveness and practical applicability of these models are often hindered by issues related to bias, lack of transparency and interpretability in their operations.

Although there is a remarkable progress in predicting liver disease with machine learning, certain challenges remain apparent. The existing models are not being used or applied in medical sectors because of lack of transparency, trustworthiness, and practical usability in a

clinical setting. Inadequate interpretability and class imbalance is a major problem as it leads to lack of trust among healthcare professionals because most of these models cannot be understood (Ganie et al., 2024; Chicco and Juman, 2021). This research is based on the development of an explainable Artificial Intelligence using machine learning technique.

1.1 RELATED WORKS

Ganie et al. (2024) conducted a study to evaluate eleven machine learning approaches for predicting liver disease from clinical data. The researchers utilized a large dataset comprising 30,691 samples with 11 features related to liver disease. They evaluated nine different algorithms, including Support Vector Machine (SVM), Logistic Regression (LR), Random Forest (RF), eXtreme Gradient Boosting (XGBoost), AdaBoost, K-Nearest Neighbors (K-NN), Decision Tree, Extra Trees, and Logistic LightGBM. XGBoost achieved the highest accuracy of 98.50%, the limitation is lack of interpretability.

Davide & Jurman (2021) explored an ensemble learning approach to enhance the classification of patients with hepatitis and cirrhosis using electronic health records (EHRs). The study analyzed EHRs from 540 healthy controls and 75 patients diagnosed with hepatitis C. The results demonstrated that the ensemble learning model, particularly using Random Forest, outperformed the traditional methods. A similar research was conducted by **Naseem et al. (2020)**

Nahar et al. (2019) conducted a comparative analysis of several ensemble techniques for liver disease prediction, specifically focusing on AdaBoost, LogitBoost, Random Forest (RF), and bagging methods using J48 and Reptree. LogitBoost achieved the highest accuracy of 71.53% among the models tested.

Kuzhippallil et al. (2020) conducted a comparative analysis of several ML techniques to predict liver disease among Indian patients, employing models such as Logistic Regression (LR), k-Nearest Neighbors (kNN), Decision Trees (DT), Random Forest (RF), Gradient Boosting

(GB), AdaBoost, XGBoost (XGB), LightGBM (LGBM), and a stacking model. LGBM achieved an accuracy of 86%, demonstrating their potential for effective liver disease prediction. The limitation is that the study did not address model interpretability, a critical factor in clinical decision-making. Other authors like Akter, L. (2022), Afrin et al. (2021), Ghosh et al. (2021) worked on early detection of liver disease using machine learning techniques.

Naseem et al. (2020) conducted a performance assessment of various classification algorithms to facilitate the early detection of liver syndrome. In their study, Naseem et al. evaluated ten different classifiers: A1DE, Multi-Layer Perceptron (MLP), Naive Bayes (NB), k-Nearest Neighbors (kNN), Support Vector Machine (SVM), CHIRP, CDT, Forest-PA, J48, and Random Forest (RF). RF exhibited a notable and highest performance with an accuracy of 72.17%.

Kalaiselvi et al,(2019) explored the use of machine learning algorithms to predict liver disease, employing methods such as k-Nearest Neighbors (kNN), Decision Trees (DT), and Adaptive Neuro-Fuzzy Inference System (ANFIS). Among the models tested, ANFIS demonstrated superior performance across all evaluated metrics.

Pasha et al. (2022) investigated the effectiveness of various machine learning (ML) techniques, specifically Random Forest, Logistic Regression, and Support Vector Machine (SVM), in predicting liver disease. The results indicated that Random Forest outperformed the other models in terms of accuracy.

Mutlu et al. (2022) conducted a study on liver disease prediction using deep learning techniques, specifically focusing on a Convolutional Neural Network (CNN) model. The study compared the CNN's performance against traditional machine learning algorithms, including Naive Bayes (NB), Support Vector Machine (SVM), K-nearest Neighbors (KNN), and Logistic Regression (LR), using two datasets: BUPA and ILPD. The CNN model achieved the highest accuracy of 75.55% on the BUPA dataset and 72.00% on the ILPD dataset. Its draw back is lack of explainable AI.

Bayani et al. (2022) conducted a study that focused on identifying predictors of esophageal varices (EV) grading in patients with cirrhosis using ensemble learning techniques, specifically CatBoost and XGBoost (XGB) classifiers. Utilizing a dataset comprising 490 patients. The findings revealed that while the CatBoost model achieved 100% precision in predicting all targets.

Dalal et al. (2022) presented a hybrid XGBoost model enhanced through hyperparameter tuning for the prediction of liver disease. The dataset comprised 416 individuals with liver problems and 167 healthy controls**,** collected from Andhra Pradesh, India. The hybrid XGBoost model significantly outperformed most conventional methods with an impressive accuracy of 93.65.

Edeh et al. (2022), examined multiple machine learning approaches to construct classification models, ultimately developing an ensemble model that combines the strengths of different algorithms. Ultimately, the overall accuracy of the ensemble model improved to 95.59. Other researchers like **Orhanbulucu et al. (2022); PredictionEdeh et al. (2022); Meng et al. (2022)** explored the use ensemble machine learning technique for detection of liver disease.

Amin et al. (2023), investigated the prediction of chronic liver disease patients using Decision Trees, Neural Networks, K-Nearest Neighbors (KNN),Random Forest and Gradient Boosting. The Random Forest classifier achieved the highest performance metrics with an accuracy; precision; recall; F1 score of 88.1%; 85.33; 92.3; 88.68 respectively. Authors like **Dritsas and Trigka (2023**), **Md et al. (2023), Mohammad et al.(2024)** also conducted research on the development of a more efficient liver disease detection approach using machine learming.

The identified gaps in existing researches are lack of external validation (Ganie et al. (2024); Kuzhippallil et al. (2020)), Model tranparency and interpretability (Ganie et al. (2024);Ghosh et al. (2021)). Small sample sizes (Davide & Jurman (2021), class imbalance (Chicco & urman (2021); Kuzhippallil et al. (2020); Kalaiselvi et al. (2019)).

2.0 METHOD

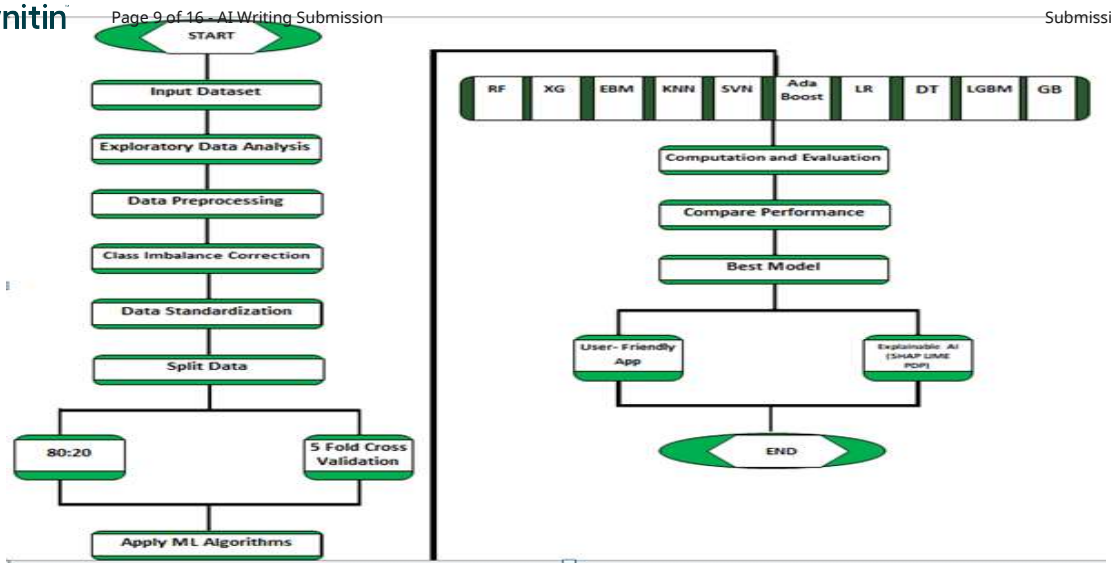
2.1 DATASET DESCRIPTION.

Datasets from the Kaggle repository was used in this project. The dataset include liver function test results and patient clinical data such as demographics and medical history. The dataset features comprised Age, Gender, BMI, Alcohol Consumption, Smoking, Genetic Risk, Physical Activity, Diabetes, Hypertension, Liver Function Test, and Diagnosis. The total number of instances in the dataset was 1,700, with 936 being not affected and 764 affected.

The design phase involved creating a detailed architecture for the liver disease prediction model . This included Exploratory Data Analysis, Data Preprocessing, Strategy to handle Class Imbalance, Data Standardization, outlining the machine learning algorithms, the integration of Explainable AI techniques and graphical User Interface to enhance interpretability. Fig 3.4 shows the block diagram of model architecture.

Fig 2.1. Block Diagram of the Architecture of Liver Disease Prediction Model

The data was first imported, an exploratory data analysis was carried out on the data. The data was preprocessed. The class imbalance strategy was applied. The dataset was Standardized and Machine learning algorithms were trained. The machine learning algorithm are Random Forest (RF), Extreme Gradient Boost (XGBoost), Light Gradient Boosting Machine (Light GBM), Gradient Boosting, Decision Tree, K Nearest Neighbor, Support Vector Machine, Ada Boost, Explainable Boosting Machine and Logistic Regression



2.2 Data Flow Diagram of the Liver Disease

3.0 RESULT

These software programs were selected for their applicability to ML workflows and their capability of working with large datasets. The liver disease prediction model was developed with tools likeWindows 10, Python 3.8, Jupyter Notebook,Anaconda, libraries and frameworks like Pandas, Numpy, Scikit Learn, MatplotLib and seaborn, Shap, PDP, Lime, Das. Also, The hardware technologies utilized in developing the system are Intel Core15Processor, 16GB Ram, 500GB SSD Storage.

3.1 PERFORMANCE TEST AND CROSS VALIDATION METRICS EVALUATION

The performance of the models were evaluated using metrics such as accuracy, precision, recall, F1-score,Confusion Matrix and AUC-ROC: . All the results of each the algorithms are presented below

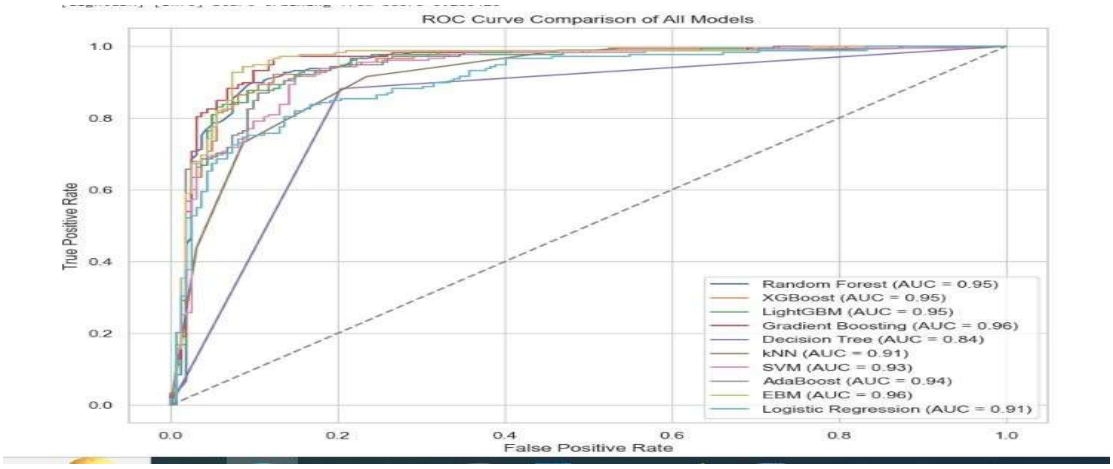


Fig 3.1 ROC Curve and AUC for all Algorithms

From the analysis, EBM and Gradient Boosting emerged as the best-performing models with AUC values of 0.96, indicating superior classification ability. Random Forest, XGBoost, and LightGBM followed closely with AUC values of 0.95, demonstrating strong predictive performance. SVM and AdaBoost performed well, while kNN, Logistic Regression, and Decision Tree had lower AUC scores, suggesting relatively weaker classification capability Overall, EBM emerged as the best model for liver disease prediction, while Decision Tree was the least effective. The confusion matrix for all the machine learning techniques is depicted in figure 4.2, Table 4.1 shows the classification performance of all machine learning the machine learning models. The ensemble models, particularly **EBM, Gradient Boosting, Random Forest, XGBoost, and LightGBM**, were the most effective for liver disease prediction. They offer high classification performance with well-balanced true positive (TP) and true negative(TN) rates. On the other hand, **Decision Tree and Logistic Regression** performed the weakest, with lower AUC values and a higher number of misclassifications. Overall, **Explainable Boosting Machine (EBM)** emerged as the best model, offering the best balance of sensitivity and specificity. With low false negatives (FN) and false positives (FP), it proved to be the most reliable for medical diagnosis. On the other hand, **the Decision Tree model** performed

the weakest, with the highest misclassification rate.

Model	TP	TN	FP	FN	AUC
EBM	168	147	15	10	**0.96**
Gradient Boosting	166	143	19	12	**0.96**
Random Forest	165	139	23	13	**0.95**
XGBoost	164	139	23	14	**0.95**
LightGBM	164	137	25	14	**0.95**
SVM	168	130	32	10	0.93
AdaBoost	163	137	25	15	0.94
kNN	163	124	38	15	0.91
Logistic Regression	152	124	38	26	0.91
Decision Tree	157	129	33	21	**0.84**

Table 3.1: Classification Performance of All Models

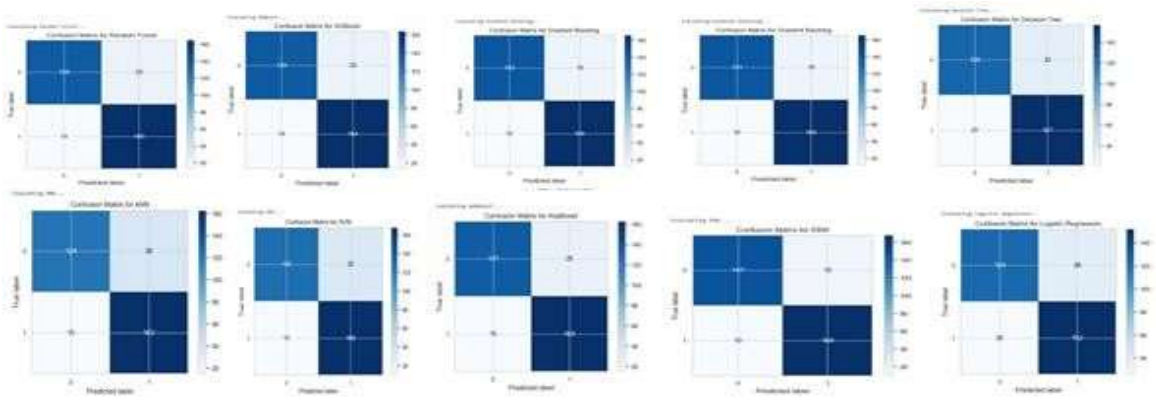


Fig 3.2: Confusion Matrix For All Algorithms

Table 3.2 shows test set results demonstrated that **EBM was the best-performing model**, achieving the highest accuracy (92.65%), precision (91.80%), recall (94.38%), specificity (90.74%), and F1- score (93.07%). This indicates that EBM maintained a strong balance between sensitivity and specificity. **Gradient Boosting (90.88%) and Random Forest (89.41%) also performed well**, with high accuracy and stable classification metrics. **XGBoost (89.12%) and LightGBM (88.53%) followed closely**, maintaining competitive precision and recall. On the other hand, **Logistic Regression (81.18%) and kNN (84.41%) showed weaker performance**, suggesting challenges with handling complex data patterns and noise.

Test Set Performance					
ALGORITHMS	ACCURACY	PRECISION	RECALL	SPECIFICITY	F1 SCORE
Random Forest	0.894118	0.877660	0.926966	0.858025	0.901639
XG Boost	0.891176	0.877005	0.921348	0.858025	0.898630
LightGBM	0.885294	0.867725	0.921348	0.845679	0.893733
Gradient Boosting	0.908824	0.897297	0.932584	0.882716	0.914601
Decision Tree	0.841176	0.826316	0.882022	0.796296	0.853261
kNN	0.844118	0.810945	0.915730	0.765432	0.860158
SVM	0.876471	0.840000	0.943820	0.802469	0.888889
Adaboost	0.882353	0.867021	0.915730	0.845679	0.890710
EBM	0.926471	0.918033	0.943820	0.907407	0.930748
Logistic Regression	0.811765	0.800000	0.853933	0.765432	0.826087

Table 3.2: Table for Test Scores for all Algorithms

Table 3.3: Cross Validation Scores for all Algorithms

Performance Matrix CV					
ALGORITHMS	ACCURACY	PRECISION	RECALL	SPECIFICITY	F1 SCORE
Random Forest	0.887500	0.893368	0.906332	0.863787	0.899804
XG Boost	0.886029	0.895151	0.901005	0.867110	0.898093
LightGBM	0.891912	0.903567	0.902375	0.878738	0.902970
Gradient Boosting	0.905147	0.908973	0.922164	0.883721	0.915521
Decision Tree	0.820588	0.853022	0.819261	0.822259	0.835801
kNN	0.806618	0.815287	0.844327	0.759136	0.829553
SVM	0.832353	0.838010	0.866755	0.789037	0.852140
Adaboost	0.883824	0.887597	0.906332	0.855482	0.896867
EBM	0.925000	0.925974	0.940633	0.905316	0.933246
Logistic Regression	0.826471	0.835476	0.857520	0.787375	0.846354

The cross-validation results shown on table 4.3 confirmed that **EBM remained the strongest model**, achieving a CV accuracy of 92.5%, reinforcing its generalization ability. **Gradient Boosting (90.51%) also performed well**, maintaining a balanced performance across different splits. **Random Forest (88.75%), XGBoost (88.60%), and LightGBM (89.19%) showed strong consistency**, aligning with their test set results. In contrast, **Decision Tree (82.05%) and kNN (80.66%) had lower CV accuracy**, reflecting their limitations in generalizing across different training subsets. **SVM (83.23%) and Logistic Regression (82.65%) showed moderate performance**, but their recall and specificity remained lower than ensemble-based

models.

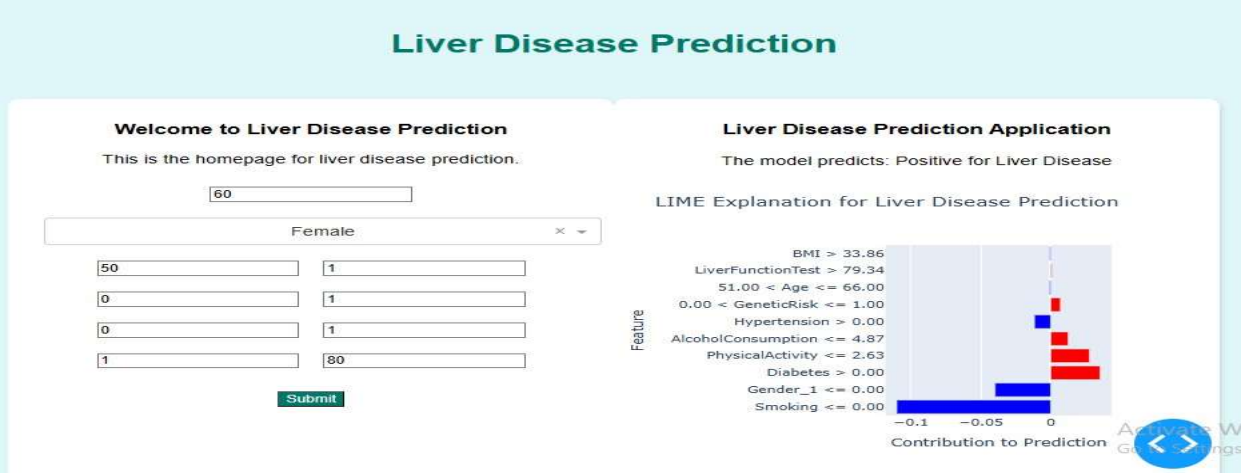
Test Set Performance Versus Cross Validation Metrics										
ALGORITHMS	ACCURACY		PRECISION		RECALL		SPECIFICITY		F1 SCORE	
	TEST	CV	TEST	CV	TEST	CV	TEST	CV	TEST	CV
Random Forest	0.894118	0.887500	0.877660	0.893368	0.926966	0.906332	0.858025	0.863787	0.901639	0.899804
XG Boost	0.891176	0.886029	0.877005	0.895151	0.921348	0.901005	0.858025	0.867110	0.898630	0.898093
LightGBM	0.885294	0.891912	0.867725	0.903567	0.921348	0.902375	0.845679	0.878738	0.893733	0.902970
Gradient Boosting	0.908824	0.905147	0.897297	0.908973	0.932584	0.922164	0.882716	0.883721	0.914601	0.915521
Decision Tree	0.841176	0.820588	0.826316	0.853022	0.882022	0.819261	0.796296	0.822259	0.853261	0.835801
kNN	0.844118	0.806618	0.810945	0.815287	0.915730	0.844327	0.765432	0.759136	0.860158	0.829553
SVM	0.876471	0.832353	0.840000	0.838010	0.943820	0.866755	0.802469	0.789037	0.888889	0.852140
Adaboost	0.882353	0.883824	0.867021	0.887597	0.915730	0.906332	0.845679	0.855482	0.890710	0.896867
EBM	0.926471	0.925000	0.918033	0.925974	0.943820	0.940633	0.907407	0.905316	0.930748	0.933246
Logistic Regression	0.811765	0.826471	0.800000	0.835476	0.853933	0.857520	0.765432	0.787375	0.826087	0.846354

Table 3.4: Comparison of Test and CV for the Ten Algorithms

A comparison of test and CV results showed that EBM consistently outperformed other models with minimal variation between the two evaluations. Gradient Boosting, XGBoost, and LightGBM showed slight drops in test accuracy, which is expected when models encounter unseen data. Random Forest demonstrated strong consistency, reinforcing its robustness. Decision Tree and kNN improved slightly in test recall but had lower specificity, indicating potential overfitting during cross-validation. SVM and Logistic Regression displayed mixed results, with Logistic Regression showing a more noticeable drop in test accuracy compared to its CV score. The top models showed minimal differences (±1%) between test and CV accuracy, demonstrating strong generalization. SVM, kNN, and Decision Tree had larger test score increases, suggesting that they were slightly underestimated in cross-validation. LightGBM and Logistic Regression experienced slight drops, though the differences were not substantial.

3.2 LIVER DISEASE PREDICTION APPLICATION

The liver disease Prediction Application was developed for real time prediction and result visualization. The App displayed both predictions and interpretability for enhanced usability and transparency. It is an intuitive interface that allowed users to input patient information like **Age**, **Gender**, and key risk factors which are **Alcohol Consumption**, **BMI**, **Hypertension, Physical**



Activity, Genetics, Smoking, Liver Function Test, and Diabetes. After submitting the data, the application predicted whether the patient has liver disease . It displayed the prediction probabilities for **Positive for Liver Disease** and **Negative for No Liver Disease** alongside with the explanations derived from explainable AI techniques. Screen Shots from the Liver Disease Prediction App is depicted in figure 4.3

Fig 4.3 : Liver Disease Prediction App with LIME Explanation.

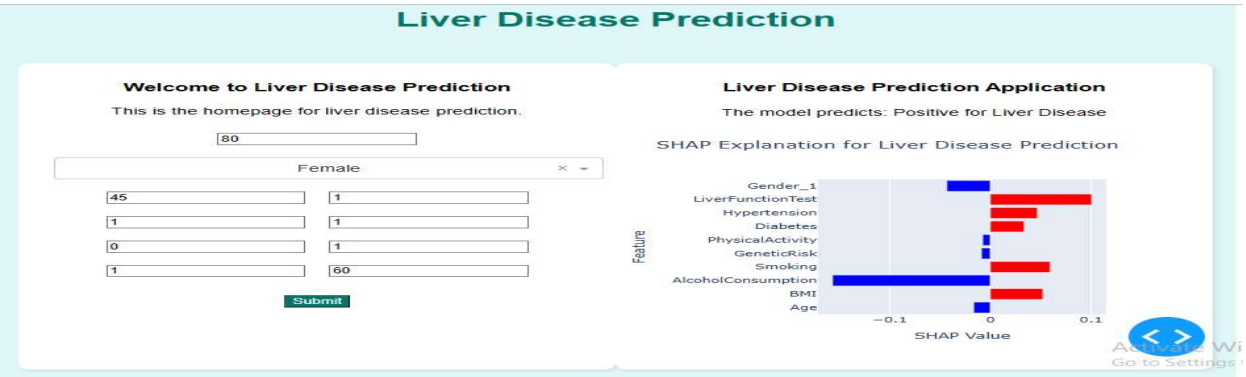


Fig 4.4 : Liver Disease Prediction App with SHAP Explanation

In this version **(Fig 4.4)**, SHapley Additive exPlanations (SHAP) replaces LIME for model explainability. The right panel now includes a SHAP value plot, detailing how each feature affects the prediction. Alcohol consumption, BMI, and smoking appear to play significant roles. The SHAP method provides a more comprehensive global and local interpretation of feature influence.

4.0 DISCUSSIONS.

The findings from this study demonstrated that machine learning can effectively predict liver disease with high accuracy while providing interpretable results through XAI techniques. The Explainable Boosting Machine (EBM) was the best-performing model. The use of Class Imbalance correction strategies significantly enhanced model performance by ensuring that minority classes were adequately represented in predictions. The implementation of Random Oversampling generated an equalized dataset, substantially improving the recall rates for minority classes without compromising the overall accuracy. Random Oversampling successfully removed bias in the liver prediction model, as recall and specificity were well-balanced. The evaluation of the test results and the cross validation results showed that the model performed consistently well.

5.0 CONCLUSIONS

This study successfully developed an XAI-integrated machine learning model for liver disease prediction, addressing key challenges such as interpretability, class imbalance, and usability in clinical practice. The Explainable Boosting Machine (EBM) was identified as the best-performing model. This study bridges the gap between advanced machine learning models and clinical applicability by providing an interpretable, user friendly and reliable liver disease prediction tool.

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